Silhouette coefficient calculation.

point	Cluster label
P_1	1
P_2	1
P_3	2
P_4	2

Dissimilarity matrix.

point	P1	P2	P3	P4
P_1	0	0.10	0.65	0.55
P_2	0.10	0	0.70	0.60
P_3	0.65	0.70	0	0.30
P_4	0.55	0.60	0.30	0

P_1 & P_2 belong to cluster 1

P_3 & P_4 belong to cluster 2

if $a<b$, $S = 1 - \left(\frac{a}{b}\right)$

if $a>b$, $S = \left(\frac{b}{a}\right) - 1$

if $a=b$, $S = 0$.

~~point pair~~

Here, a indicates average distance of a point to other points in same cluster.

b indicates average distance of a point to another point in another cluster.

point p1:- $(P_1 \rightarrow P_2) \rightarrow 0.1$, only 1 point

$$a = \frac{0.1}{1} = 0.1$$

$$\begin{matrix} P_1 \rightarrow P_3 & (0.65) \\ P_1 \rightarrow P_4 & (0.55) \end{matrix} \quad b = \frac{0.65 + 0.55}{2} = 0.6$$

$a<b$

$$SC = 1 - \left(\frac{a}{b}\right) = 1 - \frac{0.1}{0.6} = \frac{5}{6} = 0.833$$

Point p2:-

$a = (P_2 \to P_1) \to \frac{0.1}{1} \to 0.1.$

$b = \quad P_2 - P_3 \to 0.7$

$\quad\quad P_2 - P_4 \to 0.6 \quad \to \frac{0.7+0.6}{2} = 0.65$

$a<b \quad \Rightarrow \quad 1 - \frac{0.1}{0.65} = 0.84.$

$SC = 0.84.$

Point P3:-

$a = (P_3 - P_4) \to \frac{0.30}{1} = 0.30$

$b = (P_3 - P_1) \quad -0.65$

$\quad\quad (P_3 - P_2) \quad -0.70 \quad \frac{0.65+0.70}{2} = 0.675$

$a<b \Rightarrow 1 - \frac{0.3}{0.675} \Rightarrow 0.55.$

point P4:-

$a = P4 \rightarrow P3 \rightarrow \frac{0.3}{1} = 0.3$

$b = P4 \rightarrow P1 \rightarrow 0.55 \rightarrow \frac{0.55+0.60}{2}$

$P4 \rightarrow P2 \quad 0.60$

$= 0.575.$

$a<b \Rightarrow 1 - \frac{0.3}{0.575} \Rightarrow 0.478.$

point P1 SC = 0.833

point P2 SC = 0.846

point P3 SC = 0.556

point P4 SC = 0.478.

Cluster 1 Avg SC $= \frac{0.833+0.846}{2}$

$= 0.84$

Cluster 2 Avg SC $= \frac{0.556+0.478}{2}$

$= 0.517.$

Over all

Average SC $= \frac{0.840+0.517}{2} = \boxed{0.68\%}$